

Please write clearly in block capitals.

Centre number

Candidate number

Surname

Forename(s)

A-level

ENVIRONMENTAL SCIENCE**Paper 1 (Generated practice paper - Set 28)**

Assessed: The physical environment, Energy resources, Pollution, Research methods

Time allowed: 3 hours**Materials**

For this paper you may use:

- a calculator.

Instructions

- Use black ink or black ball-point pen. Pencil should only be used for drawing.
- Fill in the boxes at the top of this page.
- Answer **all** questions 1 to 10 and **one** essay from question 11.
- You must answer the questions in the spaces provided. Do not write outside the box around each page or on blank pages.
- Do all rough work in this book. Cross through any work you do not want to be marked.

Information

- The marks for questions are shown in brackets.
- The maximum mark for this paper is 120.
- All questions should be answered in continuous prose.
- You will be assessed on your ability to:
 - use good English
 - organise information clearly
 - use specialist vocabulary where appropriate.

For Examiner's Use

Question	Mark
1	
2	
3	
4	
5	
6	
7	
8	
9	
10	
11	
TOTAL	

Do not write outside the box

Answer all questions in the spaces provided.

0	1	.	1
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Using named examples, explain how the sources of noise pollution can be controlled by changes to design, operation and location.

[9 marks]

Extra space

0 1 . 2 State what is meant by the **Leq 57 dB** contour used around UK airports. **[1 mark]**

10

0 2 A student investigated whether the number of earthworms per m^2 changes with distance from a hedge across an arable field.

0 2 . 1 Explain why systematic sampling along a transect is more appropriate than random sampling for this investigation. **[2 marks]**

0 2 . 2

Describe how a preliminary study could be used to decide the distance between sampling sites and the number of samples at each site.

[4 marks]

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0 2 . 3

Describe how earthworms could be extracted from a 0.5 m x 0.5 m sample area and give **two** limitations of the method.

[4 marks]

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03 . 1

Explain how the control of emission **location** and emission **timing** can reduce the severity of pollution. Use examples.

[6 marks]

Emission location

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Emission timing

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03 . 2

Explain the **polluter pays** principle and suggest why it encourages pollution prevention.

[2 marks]

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03 . 3

Explain the **precautionary principle** as applied to the release of a new chemical.

[2 marks]

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10

0 4 . 1

Explain why compressed air energy storage with heat recovery can achieve over 90% efficiency.

[2 marks]

0 4 . 2

Explain why large-volume hot water stores lose heat more slowly per unit of stored energy than small ones.

[2 marks]

0 4 . 3

Explain why molten salt rather than water is used to store heat from CSP plants.

[2 marks]

0 4 . 4

Explain how high thermal mass building materials reduce energy use.

[2 marks]

0 4 . 5

Describe **one** other energy storage technology.

[2 marks]

10

0 5 . 1

Give **three** sources of unburnt hydrocarbons in urban air.

[3 marks]

1

2

3

0 5 . 2

Describe **two** methods used to control hydrocarbon emissions at petrol filling stations.

[2 marks]

1

2

0 5 . 3

Explain how a catalytic converter controls the emissions of carbon monoxide, hydrocarbons and NO_x.

[3 marks]

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0 5 . 4

Explain why carbon monoxide is toxic to humans.

[2 marks]

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0 6 . 1 Explain how each of the following features of vehicle design reduces energy use.

[6 marks]

Aerodynamics

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Reduced mass

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Regenerative braking (KERS)

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0 6 . 2 Explain why solid steel wheels on rails lose less energy than pneumatic tyres on roads.

[2 marks]

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0 6 . 3 Describe **two** ways combustion efficiency in an internal combustion engine can be improved.

[2 marks]

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1

2

10

0 7 . 1 Describe the two processes that drive the North Atlantic Conveyor. [2 marks]

0 7 . 2 Explain how global climate change could weaken the North Atlantic Conveyor and describe the possible consequence for the climate of the UK. [4 marks]

0 7 . 3 Explain why the effects of climate change on UK temperatures are difficult to predict. [3 marks]

07 . 4 Describe how deep ocean currents are monitored.

[1 mark]

10

08 The Universal Soil Loss Equation (USLE) is $A = R \times K \times LS \times C \times P$. Table 1 gives factor values for a farm.

Table 1

Factor	Description	Value
R	Rainfall erosivity for the farm	110
K	Soil erodibility: loam	0.67
LS	Slope length and gradient: 120 m, 6%	1.35
C (crop type x tillage)	Beans (0.50) with spring ploughing (0.90); or wheat (0.35) with mulch tillage (0.60)	see question
P	Ploughing up and down slope 1.0; contour ploughing 0.50; strip cropping with contour ploughing 0.25	see question

Table 2

Soil erosion class	Rate of soil loss / $\text{t ha}^{-1} \text{yr}^{-1}$
Low	under 5
Moderate	5-10
High	10-20
Very high	20-30
Severe	over 30

08 . 1 Calculate the annual soil loss for beans with spring ploughing, ploughed up and down the slope. Give your answer to one decimal place.

[2 marks]

Answer

t/ha/yr

08 . 2

Calculate the annual soil loss for wheat with mulch tillage and strip cropping with contour ploughing.

[2 marks]

Answer

t/ha/yr

08 . 3

The natural rate of soil formation on the farm is $1 \text{ t ha}^{-1} \text{ yr}^{-1}$. Use your answers and Table 2 to evaluate the sustainability of the two farming systems.

[3 marks]

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08 . 4

Explain why the USLE gives only an estimate of soil loss.

[2 marks]

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08 . 5

Suggest which single factor in the USLE a farmer can change most easily.

[1 mark]

10

0 9 A sand and gravel pit near a town is due to close. Three options for the site are proposed: a wetland nature reserve, housing, or return to farmland.

0 9 . **1** Explain why flooded sand and gravel pits are often developed as wetland nature reserves.

[2 marks]

0 9 . **2** Explain the conditions that would need to be met for the site to be used for housing or farmland.

[2 marks]

0 9 . **3** Explain why moving the wildlife from a site before mining begins is rarely completely successful.

[1 mark]

5

Neonicotinoids are the most widely used group of insecticides. Explain how the properties of neonicotinoids affect their impact on bee populations and describe the methods used to reduce this impact.

ESP/7447/1/SET28

Extra space

10 . **2** State **one** way in which pesticide use in agriculture may be reduced by using natural processes.

[1 mark]

10

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Write an essay on one of the following topics.

1	1	.	1
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Mineral resources are non-renewable. Discuss how an understanding of geological processes and the development of new technologies can extend the period for which society can rely on mineral resources.

[25 marks]

OR

1 1 . 2

Discuss how knowledge of the properties of pollutants can be used to reduce the severity of their impact on the environment.

[25 marks]

Indicate which question you have answered by writing its number, **11 . 1** or **11 . 2**, at the top of your answer.

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END OF QUESTIONS